MD. ASHIQUR RAHMAN KHAN

Data Analyst | Public Health and Healthcare Analytics | B.Sc. in Statistics

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LinkedIn | GitHub | Portfolio

Summary

Statistics graduate from MBSTU working in public health and healthcare analytics. Built end-to-end projects on national health survey data and hospital readmission using survival analysis, Cox regression, competing risks, causal inference and machine learning. Works in R, Python, SQL and Power BI, from data cleaning and survey-weighted modeling to dashboards and written reports. Seeking a Data Analyst role in public health or healthcare, with a long-term goal of graduate study in Data Science and Machine Learning.

Projects

Cardiometabolic Risk and Mortality in US Adults (NHANES)
2026 | *GitHub | Python, DuckDB SQL, R, Power BI*

- Followed 50,819 US adults from NHANES 1999-2018 linked to National Death Index mortality (8,279 deaths, 479,971 person-years) to measure how diabetes, hypertension, obesity and chronic kidney disease (CKD) raise the risk of death
- Fitted survey-weighted Cox models over 20 multiple imputations: diabetes HR 1.44, CKD HR 1.59, smoking HR 2.27; CKD explained 14.7% of deaths
- Showed the crude prediabetes excess was age (adjusted HR 0.99), and that 1 minus Kaplan-Meier overstated cause-specific risk by up to 28.5% against the Aalen-Johansen competing-risks estimator
- Estimated total effects with inverse probability weighting and E-values, and built a 10-year risk model (C-index 0.872 on later survey cycles), an 8-page Power BI dashboard and a 50-page report

Diabetes 30-Day Hospital Readmission Analysis
2026 | *GitHub | Python, DuckDB SQL, XGBoost, SHAP, Power BI*

- Analyzed 69,970 diabetic patients from 130 US hospitals (8.97% readmitted in 30 days) with 13 SQL business questions, effect-size EDA and logistic regression
- Tested whether HbA1c measurement goes with fewer readmissions: adjusted OR 0.96, propensity-matched OR 0.99, and a significant interaction with primary diagnosis ($p = 0.0067$)
- Built a calibrated XGBoost risk model with SHAP (ROC-AUC 0.659) in which the top 20% of patients by predicted risk hold 38% of readmissions, plus an 8-page Power BI dashboard with 158 DAX measures

Dunnhumby Retail Analytics
2026 | *GitHub | SQL, DuckDB, Python, Power BI*

- Analyzed 2,500 households over 2 years of grocery data with RFM segmentation, Kaplan-Meier and Cox retention analysis, and SQL market basket analysis
- Showed a naive campaign uplift of $31.40 a week fell to $1.03 (not significant) after propensity score matching and difference-in-differences

Berka Retail Banking Analytics
2026 | *GitHub | Python, DuckDB, SQL, Power BI, DAX*

- Built an ETL pipeline and star schema over 1,056,320 transactions and 4,500 accounts, feeding a 7-page Power BI dashboard with 77 DAX measures
- Corrected right-censoring bias in loan default rates: loans above 250K CZK defaulted at about 23% against about 7% under 100K

Experience

Undergraduate Researcher - Department of Statistics
2024 - 2026 | *Mawlana Bhashani Science and Technology University (MBSTU)*

- Compared SARIMA with Random Forest, XGBoost, SVR, LSTM and a hybrid model on 43 years of monthly NASA POWER wind speed data for 64 districts of Bangladesh (GitHub)
- Reached a holdout R-squared of 0.83 (RMSE 0.236 m/s); Diebold-Mariano tests and 5-fold rolling-origin cross-validation showed no model significantly better than SARIMA
- Found and fixed a data leakage error in the rolling features, and wrote the results up in LaTeX with reproducible R and Python workflows

Languages

Bengali | English | Hindi

Education

B.Sc. in Statistics
Mawlana Bhashani Science and Technology University (MBSTU)
2021 - 2026, Tangail, Bangladesh
CGPA: 3.24 / 4.00 (expected)

- Coursework: Survival Analysis, Regression, Bayesian Statistics, Time Series, Multivariate Analysis

Skills

Programming Languages
R | Python | SQL | DAX | LaTeX

Public Health and Biostatistics
Epidemiology | Biostatistics | Survival Analysis | Cox Regression | Kaplan-Meier | Competing Risks | Survey-Weighted Analysis | Multiple Imputation | Causal Inference | Propensity Score Matching | Logistic Regression | Risk Prediction

Statistics and Machine Learning
Hypothesis Testing | ANOVA | Chi-Square | Linear Regression | Bayesian Inference | Time Series | SARIMA | Machine Learning | Predictive Modeling

Libraries and Frameworks
pandas | NumPy | scikit-learn | XGBoost | SHAP | statsmodels | lifelines | scipy | matplotlib | seaborn | tidyverse | ggplot2 | survey | survival | mice | forecast

Tools and Platforms
Power BI | Excel | Tableau | DuckDB | MySQL | SPSS | RStudio | Jupyter | Google Colab | Git | GitHub

Core Competencies
Data Cleaning | EDA | ETL | Data Modeling | Data Visualization | Dashboards | Report Writing | Communication | Problem Solving

Certifications

Power BI + Claude: AI-Powered Modeling with the MCP Connector
LinkedIn Learning | *2026*

Python for Data Science and Machine Learning Essential Training (Parts 1 and 2)
LinkedIn Learning | *2026*

Bayesian Statistics: From Concept to Data Analysis
UC Santa Cruz, Coursera | *2025*

Regression Models
Johns Hopkins University, Coursera | *2025*

Basic Programming with Python
Bangladesh Computer Council (EDGE) | *2025*

Data Analytics and Power BI Career Path
Interactive Cares | *2025*

Generative AI with Large Language Models
DeepLearning.AI and AWS, Coursera | *2025*

Generative AI for Everyone
DeepLearning.AI, Coursera | *2025*

Introduction to Data Analytics
IBM, Coursera | *2024*